

# Simple Linear Regression

Grace Carmichael

Department of Statistical Sciences  
University of Cape Town

2024



# Outline

- 1 Recap of Foundational Concepts in Statistics
- 2 The Problem We Want to Solve:
- 3 Example Problem
- 4 Correlation Analysis
- 5 Simple Linear Regression

# Recap of Foundational Concepts in Statistics

# Population versus Sample

## Population



## Sample



- When we refer to a numerical descriptor for a population we refer to it as a parameter, where a numerical descriptor for a sample is referred to as a statistic.
- We use sample statistics to approximate population parameters.

# Statistical Inference

Statistical inference is the attempt to reach a conclusion concerning a complete set of observations (the **population**) using only a subset thereof (a **sample**).

- It is important to note that this sample needs to be representative of the population in order to make accurate inference.
- We make use of sampling distributions to make inference.
- Statistical inference is conducted with the help of hypothesis testing.

# Hypothesis Testing

Hypothesis testing allows us to make statements about a population from a sample of that population. It involves the following basic steps:

- Step 1: Define the null hypothesis ( $H_0$ ) This is the hypothesis of no statistical significance.
- Step 2: Define the alternative hypothesis ( $H_a$ ) This is the hypothesis of statistical significance.
- Step 3: Define the significance level ( $\alpha$ ) This is the type one error rate (probability of falsely rejecting  $H_0$ ). Typically  $\alpha = 0.05$  or  $\alpha = 0.01$  are sufficiently low.
- Step 4: Calculate the test statistic This will be calculated differently depending on the test being conducted.
- Step 5: Find the p-value This is the probability of getting a result as or more extreme than the observed test statistic, assuming  $H_0$  is true. A precise p-value can be generated using software or an approximate one using tables by hand.
- Step 6: Make a conclusion If p-value is  $\leq \alpha$ , then we reject  $H_0$  and conclude statistical significance of our result. Otherwise, we fail to reject  $H_0$  and conclude no statistical significance (this means that we can't make any statements about the population from our sample result).

# The Problem We Want to Solve:

# Describing the relationship between two variables

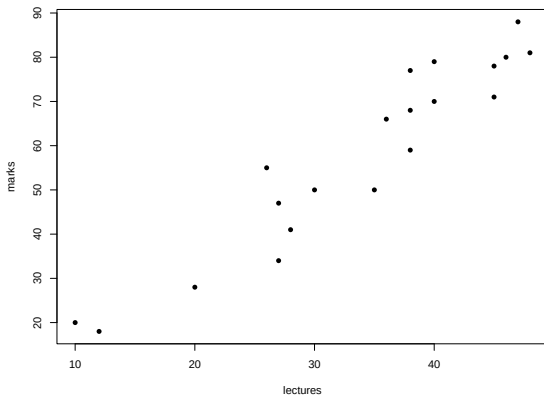
- How strong is the relationship? (so we want to be able to quantify it)
- Is this observed relationship likely real or just due to chance?
- Can we explain the impact that changing one variable has on another variable?
- Can we predict the value of one variable from another variable?

# Example Problem

# Lecture attendance example

As part of an experiment, a lecturer recorded the overall course marks and number of lectures attended for 20 students in the course that they teach. The results of this experiment are shown below:

|    | Number of lectures | Marks |
|----|--------------------|-------|
| 1  | 46                 | 80    |
| 2  | 10                 | 20    |
| 3  | 38                 | 59    |
| 4  | 27                 | 34    |
| 5  | 45                 | 71    |
| 6  | 26                 | 55    |
| 7  | 35                 | 50    |
| 8  | 45                 | 78    |
| 9  | 48                 | 81    |
| 10 | 20                 | 28    |
| 11 | 30                 | 50    |
| 12 | 27                 | 47    |
| 13 | 38                 | 77    |
| 14 | 12                 | 18    |
| 15 | 28                 | 41    |
| 16 | 40                 | 79    |
| 17 | 38                 | 68    |
| 18 | 47                 | 88    |
| 19 | 36                 | 66    |
| 20 | 40                 | 70    |



# Correlation Analysis

# Correlation Analysis as a method to solve our problem

Correlation is a measure of strength and direction of a linear relationship between two variables.

- Correlation is bounded between -1 and 1.
- Correlation does not have a unit.
- Correlation can not be used to predict one variable from another.

# Correlation coefficient

Correlation is measured using the correlation coefficient (typically the Pearson correlation coefficient).

- The **population correlation coefficient** ( $\rho$ ) measures the direction and strength of the association between the full set of two variables.
- The **sample correlation coefficient** ( $r$ ) is an estimate of  $\rho$  and measures the direction and strength of the association between the two variables in a sample of the population.

The sample correlation coefficient is given by:

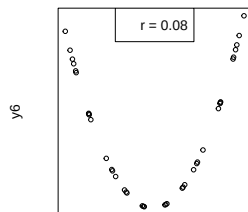
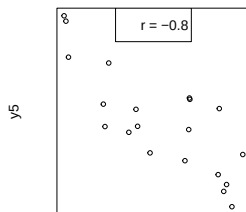
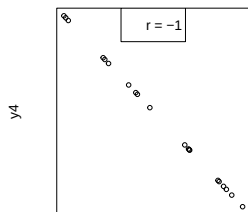
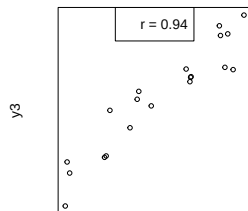
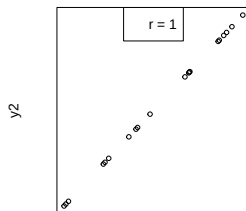
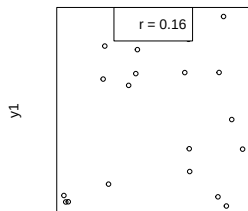
$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

# Test your understanding

Calculate the correlation coefficient between **X** and **Y** for the following 3 observations:

| <b>X</b> | <b>Y</b> |
|----------|----------|
| 2        | 3        |
| 3        | 1        |
| 4        | 2        |

# Examples of data with different correlation coefficients



# Correlation Analysis with our Example

|    | Number of lectures | Marks |
|----|--------------------|-------|
| 1  | 46                 | 80    |
| 2  | 10                 | 20    |
| 3  | 38                 | 59    |
| 4  | 27                 | 34    |
| 5  | 45                 | 71    |
| 6  | 26                 | 55    |
| 7  | 35                 | 50    |
| 8  | 45                 | 78    |
| 9  | 48                 | 81    |
| 10 | 20                 | 28    |
| 11 | 30                 | 50    |
| 12 | 27                 | 47    |
| 13 | 38                 | 77    |
| 14 | 12                 | 18    |
| 15 | 28                 | 41    |
| 16 | 40                 | 79    |
| 17 | 38                 | 68    |
| 18 | 47                 | 88    |
| 19 | 36                 | 66    |
| 20 | 40                 | 70    |

$$\bar{x} = 33.8 \quad \text{and} \quad \bar{y} = 58$$

$$r = \frac{SS_{xy}}{\sqrt{SS_x SS_y}} = \frac{4280}{\sqrt{2345.2 \times 8660}} = 0.95$$

# Example in R

```
#-----
# Perform Correlation Analysis on example data using first principals
#-----
```

```
x <- lectures
y <- marks

# calculate means of x and y
xbar <- mean(x)
ybar <- mean(y)

# calculate sum of squares
SSxy <- sum((x-xbar)*(y-ybar))
SSx <- sum((x-xbar)^2)
SSy <- sum((y-ybar)^2)

# calculate correlation
(r <- SSxy/sqrt(SSx*SSy))
```

```
## [1] 0.9497185
```

```
#-----
# Confirm result using base R cor() function
#-----
# cor() just takes the variables as inputs and outputs the correlation
cor(x,y)
```

```
## [1] 0.9497185
```

# Inference on correlation coefficient (is it significant?)

Step 1: Define the null hypothesis  $H_0: \rho = 0$

Step 2: Define the alternative hypothesis  $H_a: \rho \neq 0$

Step 3: Define the significance level  $\alpha = 0.05$

Step 4: Calculate the test statistic

$$tstat = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2} \quad tstat = \frac{0.9497185 \times \sqrt{20-2}}{\sqrt{1-0.9497185^2}}$$

$$tstat = 12.87 \sim t_{18}$$

## Step 5: Find the p-value

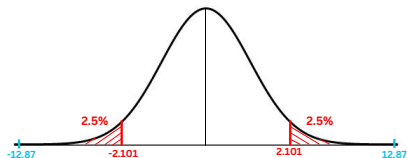


TABLE 2. *t*-DISTRIBUTION: One sided critical values, i.e. the value of  $t_{df}^P$  such that  $P = \Pr[t_{df} > t_{df}^P]$ , where  $df$  is the degrees of freedom

| df | Probability Level P |       |       |       |       |       |       |       |        |       |
|----|---------------------|-------|-------|-------|-------|-------|-------|-------|--------|-------|
|    | 0.4                 | 0.3   | 0.2   | 0.1   | 0.05  | 0.025 | 0.01  | 0.005 | 0.0025 | 0.001 |
| 1  | 0.325               | 0.727 | 1.376 | 3.078 | 6.314 | 12.71 | 31.82 | 63.66 | 127.3  | 318.3 |
| 2  | 0.289               | 0.617 | 1.061 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 14.09  | 22.33 |
| 3  | 0.277               | 0.584 | 0.978 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 7.453  | 10.21 |
| 4  | 0.271               | 0.569 | 0.941 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 5.598  | 7.173 |
| 5  | 0.267               | 0.559 | 0.920 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 4.773  | 5.894 |
| 6  | 0.265               | 0.553 | 0.906 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 4.317  | 5.208 |
| 7  | 0.263               | 0.549 | 0.896 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.029  | 4.785 |
| 8  | 0.262               | 0.546 | 0.889 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 3.833  | 4.501 |
| 9  | 0.261               | 0.543 | 0.883 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 3.690  | 4.297 |
| 10 | 0.260               | 0.542 | 0.879 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 3.581  | 4.144 |
| 11 | 0.260               | 0.540 | 0.876 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 3.497  | 4.025 |
| 12 | 0.259               | 0.539 | 0.873 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.428  | 3.930 |
| 13 | 0.259               | 0.538 | 0.870 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.372  | 3.852 |
| 14 | 0.258               | 0.537 | 0.868 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.326  | 3.787 |
| 15 | 0.258               | 0.536 | 0.866 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.286  | 3.733 |
| 16 | 0.258               | 0.535 | 0.865 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.252  | 3.686 |
| 17 | 0.257               | 0.534 | 0.863 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.222  | 3.646 |
| 18 | 0.257               | 0.534 | 0.862 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.197  | 3.610 |
| 19 | 0.257               | 0.533 | 0.861 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.174  | 3.579 |
| 20 | 0.257               | 0.533 | 0.860 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.153  | 3.552 |

```
#-----
# Perform hypothesis test on correlation
#-----
```

```
# Using the t-stat that we calculated by hand
2*pt(q=12.87,df=18, lower.tail = F)
```

```
## [1] 1.622399e-10
```

```
# using cor.test() which conducts the entire hypothesis test for you
cor.test(x,y)
```

```
##
## Pearson's product-moment correlation
##
## data: x and y
## t = 12.869, df = 18, p-value = 1.625e-10
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.8748863 0.9802637
## sample estimates:
## cor
## 0.9497185
```

Step 6: Make a conclusion the p-value is sufficiently small to reject  $H_0$  (p-value  $< 0.05$ ). Thus, it can be concluded that there is some significant linear relationship between course marks and the number of lectures attended by the students.

# Limitations of correlation analysis

Recall the questions we want to answer about our two variables of interest:

- How strong is the relationship? (so we want to be able to quantify it)
- Is this observed relationship likely real or just due to chance?
- Can we explain the impact that changing one variable has on another variable?
- Can we predict the value of one variable from another variable?

Correlation analysis allows us to answer the first two questions but the limitation of correlation analysis is the inability to quantify the impact of one variable on another and to predict the value of one variable from the other.

# Simple Linear Regression

# Simple Linear Regression as a method to solve our problem

Simple linear regression is another method used to describe the relationship between two variables. It is different to correlation analysis in that:

- There is a **dependent** and **independent** variable. The dependent variable is the one we want to explain and the independent variable is the one we want to use to explain the dependent variable.
- It allows us to explain the impact of changing the independent variable on the dependent variable.
- It allows us to predict the dependent variable from the independent variable (if our model fit is good).

# Simple Linear Regression Model

$$y = mx + c$$

## Population model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Where:

- $i$  refers to a specific observation
- $\beta_0$  is the intercept parameter
- $\beta_1$  is the slope parameter
- $\epsilon_i$  is the error

The error term accounts for any variability in the response that is not due to the independent variable.

## Sample model

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Where:

- $i$  refers to a specific observation
- $\hat{y}_i$  is the predicted value of the dependent variable
- $\hat{\beta}_0$  is the estimated regression intercept
- $\hat{\beta}_1$  is the estimated regression slope

Assume that  $E[\epsilon_i] = 0$  and that  $\epsilon_i \sim N(0, \sigma^2)$

# Estimating the $\beta$ parameters

The algorithm used to find the optimal values of  $\beta_0$  and  $\beta_1$  in a simple linear regression model is called the Ordinary Least Squares (OLS) algorithm. It works by minimizing the sum of the squared error terms, ie:

$$\text{Minimize } \sum_{i=1}^n \epsilon_i^2$$

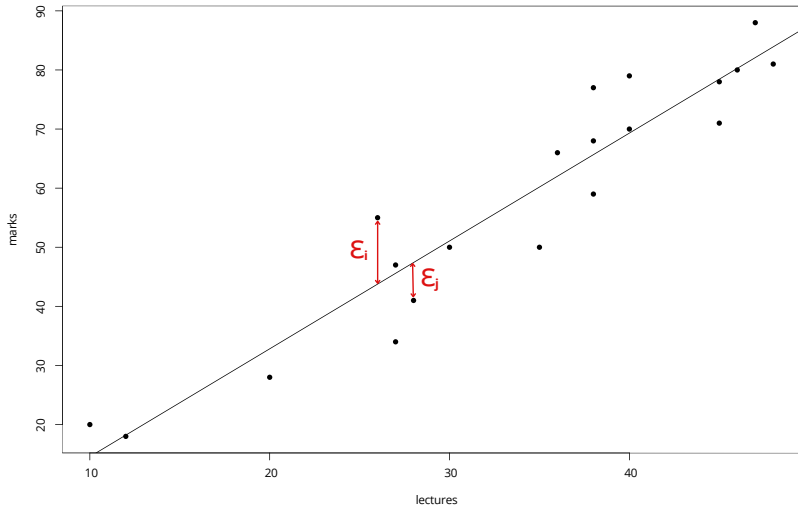
Where:

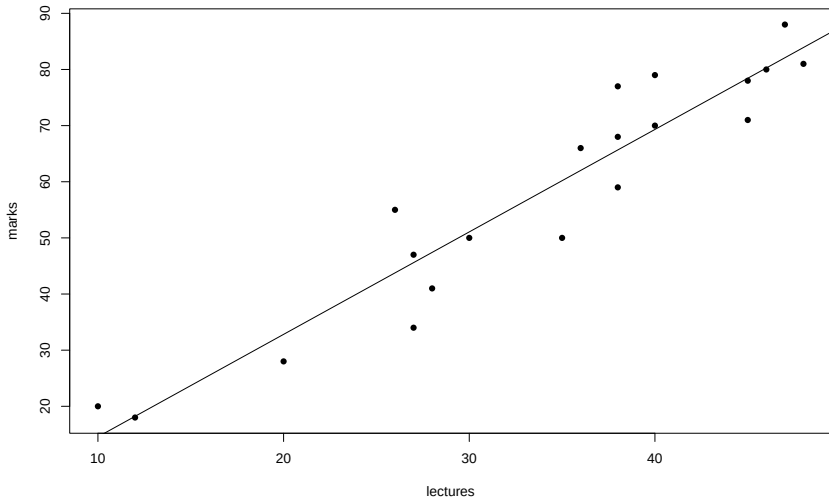
$$\epsilon_i = y_i - \hat{y}_i$$

So:

$$\sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n \left[ y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right]^2$$

We can express  $\epsilon_i$  in terms of the  $\beta$  estimates and can thus find values of the  $\beta$  estimates that minimize  $\sum_{i=1}^n \epsilon_i^2$ . These will be the optimal  $\beta$  values for the linear relationship we are trying to model.





# Performing Linear Regression in R

```
fit <- lm(marks~lectures)
summary(fit)
```

```
##
## Call:
## lm(formula = marks ~ lectures)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.590   -5.215   -0.240    4.348   11.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.6851     5.0333  -0.732   0.474
## lectures       1.8250     0.1418  12.869 1.62e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.868 on 18 degrees of freedom
## Multiple R-squared:  0.902, Adjusted R-squared:  0.8965
## F-statistic: 165.6 on 1 and 18 DF, p-value: 1.625e-10
```

- $\hat{\beta}_0 \longrightarrow$  (Intercept) estimate
- $\hat{\beta}_1 \longrightarrow$  lectures estimate

So the regression equation for the example is given by:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1(\text{lectures})$$

$$\hat{y} = -3.6851 + 1.8250(\text{lectures})$$

**Interpretation of  $\beta_0$  (intercept)** On average, a student's course mark is -3.6851 when a student attends no lectures (this is not contextually practical as a student's mark cannot be  $< 0$ ).

**Interpretation of  $\beta_1$  (slope)** On average, there is a 1.8250 mark increase in student marks for the course for each additional lecture attended.

# General interpretation of $\beta$ coefficients

**Interpretation of  $\beta_0$  (intercept)** It is the average estimated value of  $y$  when  $x = 0$ .

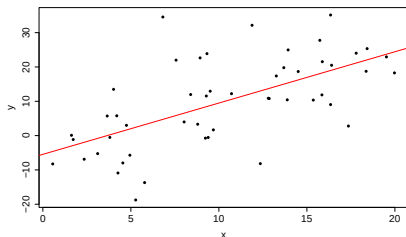
**Interpretation of  $\beta_1$  (slope)** It is the average estimated change in  $y$  for a unit increase in  $x$ . If  $\beta_1$  is positive then the change is an increase and negative means a decrease (It is important to be **specific**).

# Assessing the accuracy of the model

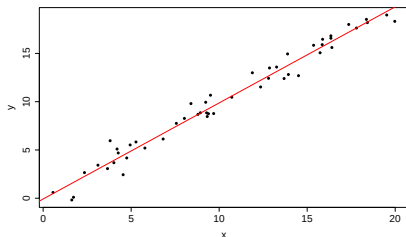
To assess the overall accuracy of the model estimates, we can use a measure called the **Residual Standard Error (RSE)**. This measures the standard deviation of the model residuals (average amount that the response will deviate from the regression line).

$$RSE = \sqrt{\frac{\sum_{i=1}^n \epsilon_i^2}{n - 2}}$$

## Higher RSE



## Lower RSE

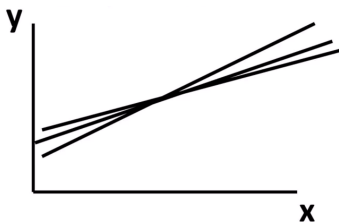


# Assessing the accuracy of the $\beta$ estimates

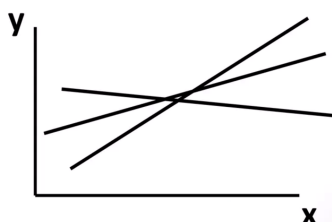
The standard error of an estimate indicates how different the population estimate is likely to be from the sample estimate. A large standard error relative to the size of the estimate is an indication that the estimate may not be an accurate reflection of the true population parameter.

$$se(\beta_1) = \frac{RSE}{\sqrt{\sum(x_i - \bar{x})^2}} = \frac{RSE}{\sqrt{SS_x}}$$

**Smaller  $se(\beta_1)$**



**Larger  $se(\beta_1)$**



Where each line on the plot is the regression line from a different sample.

# Testing the significance of $\beta_1$ estimate

We are usually mostly interested in the significance of the slope estimate ( $\beta_1$ ), as this is the estimate that tells us whether there is a relationship between the independent and dependent variables. We can do a hypothesis test on this estimate as follows:

Step 1: Define the null hypothesis  $H_0: \beta_1 = 0$

Step 2: Define the alternative hypothesis  $H_a: \beta_1 \neq 0$

Step 3: Define the significance level  $\alpha = 0.05$

Step 4: Calculate the test statistic

$$tstat = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} \sim t_{n-2} \quad tstat = \frac{1.825}{0.1418}$$

$$tstat = 12.87 \sim t_{18}$$

## Step 5: Find the p-value

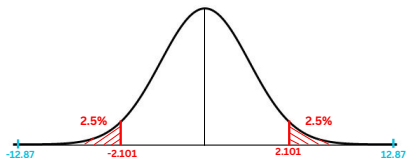


TABLE 2. *t*-DISTRIBUTION: One sided critical values, i.e. the value of  $t_{df}^P$  such that  $P = \Pr(t_{df} > t_{df}^P)$ , where  $df$  is the degrees of freedom

| $df$ | Probability Level $P$ |       |       |       |       |       |       |       |        |       |        |
|------|-----------------------|-------|-------|-------|-------|-------|-------|-------|--------|-------|--------|
|      | 0.4                   | 0.3   | 0.2   | 0.1   | 0.05  | 0.025 | 0.01  | 0.005 | 0.0025 | 0.001 | 0.0005 |
| 1    | 0.325                 | 0.727 | 1.376 | 3.078 | 6.314 | 12.71 | 31.82 | 63.66 | 127.3  | 318.3 | 636.6  |
| 2    | 0.289                 | 0.617 | 1.061 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 14.09  | 22.33 | 31.60  |
| 3    | 0.277                 | 0.584 | 0.978 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 7.453  | 10.21 | 12.92  |
| 4    | 0.271                 | 0.569 | 0.941 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 5.598  | 7.173 | 8.610  |
| 5    | 0.267                 | 0.559 | 0.920 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 4.773  | 5.894 | 6.869  |
| 6    | 0.265                 | 0.553 | 0.906 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 4.317  | 5.208 | 5.959  |
| 7    | 0.263                 | 0.549 | 0.896 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.029  | 4.785 | 5.408  |
| 8    | 0.262                 | 0.546 | 0.889 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 3.833  | 4.501 | 5.041  |
| 9    | 0.261                 | 0.543 | 0.883 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 3.690  | 4.297 | 4.781  |
| 10   | 0.260                 | 0.542 | 0.879 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 3.581  | 4.144 | 4.587  |
| 11   | 0.260                 | 0.540 | 0.876 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 3.497  | 4.025 | 4.437  |
| 12   | 0.259                 | 0.539 | 0.873 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.428  | 3.930 | 4.318  |
| 13   | 0.259                 | 0.538 | 0.870 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.372  | 3.852 | 4.221  |
| 14   | 0.258                 | 0.537 | 0.868 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.326  | 3.787 | 4.140  |
| 15   | 0.258                 | 0.536 | 0.866 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.286  | 3.733 | 4.073  |
| 16   | 0.258                 | 0.535 | 0.865 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.252  | 3.686 | 4.015  |
| 17   | 0.257                 | 0.534 | 0.863 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.222  | 3.646 | 3.965  |
| 18   | 0.257                 | 0.534 | 0.862 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.197  | 3.610 | 3.922  |
| 19   | 0.257                 | 0.533 | 0.861 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.174  | 3.579 | 3.883  |
| 20   | 0.257                 | 0.533 | 0.860 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.153  | 3.552 | 3.850  |

```

-----
# Perform hypothesis test on slope coefficient
-----

# Using the t-stat that we calculated by hand
2*pt(q=12.87,df=18, lower.tail = F)

```

```
## [1] 1.622399e-10
```

```

# using the output from lm()
fit <- lm(marks~lectures)
summary(fit)

```

```

##
## Call:
## lm(formula = marks ~ lectures)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.590   -5.215   -0.240    4.348   11.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.6851     5.0333   -0.732   0.474
## lectures       1.8250     0.1418  12.869 1.62e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.868 on 18 degrees of freedom
## Multiple R-squared:  0.902, Adjusted R-squared:  0.8965
## F-statistic: 165.6 on 1 and 18 DF,  p-value: 1.625e-10

```

Step 6: Make a conclusion the p-value is sufficiently small to reject  $H_0$  (p-value  $< 0.05$ ). Thus, it can be concluded that there is some significant linear relationship between course marks and the number of lectures attended by the students.

# Confidence Intervals for $\beta$ Estimates

We can get a range of values that we are relatively sure that the true  $\beta$  parameters fall into (with a given level of confidence).

We calculate this range using:

$$CI = \hat{\beta}_j \pm t_{\frac{\alpha}{2}, df} \times se(\hat{\beta}_j)$$

We can calculate the 95% confidence interval for  $\beta_1$  (the slope) as follows:

$$CI = \hat{\beta}_1 \pm t_{\frac{\alpha}{2}, 18} \times se(\hat{\beta}_1)$$

$$CI = 1.825 \pm 2.101 \times 0.1418$$

$$CI = [1.527, 2.123]$$

```
#-----
# Calculate confidence interval of regression estimates in R
#-----

confint(fit)

##                2.5 %    97.5 %
## (Intercept) -14.259806 6.889518
## lectures    1.527061 2.122947
```

Be careful not to interpret the confidence interval as a 95% probability that the parameter is between the interval, this is **WRONG!** It really just means that if we resample our population we expect 95% of the estimates to be within our interval.

# Checking overall model significance

We have seen that the significance of the  $\beta$  estimates can be assessed. We can also check the overall model significance by assessing if our model is significantly different to a null model (a model with just an intercept).

| Source of Variation | df      | SS                                                 | Mean Square                                         | F                                           |
|---------------------|---------|----------------------------------------------------|-----------------------------------------------------|---------------------------------------------|
| Regression          | 1       | $SS_{\text{reg}} = \sum_i (\hat{y}_i - \bar{y})^2$ | $MS_{\text{reg}} = \frac{SS_{\text{reg}}}{1}$       | $\frac{MS_{\text{reg}}}{MS_{\text{resid}}}$ |
| Residual            | $n - 2$ | $SS_{\text{resid}} = \sum_i (y_i - \hat{y}_i)^2$   | $MS_{\text{resid}} = \frac{SS_{\text{resid}}}{n-2}$ |                                             |
| Total               | $n - 1$ | $SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2$       |                                                     |                                             |

We can use the F-statistic to then perform an F-test.

Note:  $\sqrt{MS_{\text{resid}}} = RSE$

In R the test for overall model significance is automatically done when running `lm()` and the results can be viewed by summarising the model object:

```
fit <- lm(marks~lectures)
summary(fit)
```

```
##
## Call:
## lm(formula = marks ~ lectures)
##
## Residuals:
```

|  | Min     | 1Q     | Median | 3Q    | Max    |
|--|---------|--------|--------|-------|--------|
|  | -11.590 | -5.215 | -0.240 | 4.348 | 11.335 |

```
##
## Coefficients:
```

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -3.6851  | 5.0333     | -0.732  | 0.474        |
| lectures    | 1.8250   | 0.1418     | 12.869  | 1.62e-10 *** |

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.868 on 18 degrees of freedom
## Multiple R-squared:  0.902, Adjusted R-squared:  0.8965
## F-statistic: 165.6 on 1 and 18 DF, p-value: 1.625e-10
```

# Coefficient of determination ( $R^2$ )

The coefficient of determination is a measure of model fit. It is calculated using:

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} = \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}$$

- The coefficient of determination (or R-squared) is the square of the sample correlation coefficient.
- It describes the proportion of variation in the response variable that is explained by the explanatory variable.
- It is bounded between 0 and 1.
- A low value of  $R^2$  (closer to 0) indicates a poor model fit (only a small proportion of the variation in  $y$  is explained by  $x$ ) and a large value (closer to 1) indicates a good fit.

# Coefficient of determination in R

The coefficient of determination can also be found in the model summary in R.

```
fit <- lm(marks~lectures)
summary(fit)
```

```
##
## Call:
## lm(formula = marks ~ lectures)
##
## Residuals:
```

|  | Min     | 1Q     | Median | 3Q    | Max    |
|--|---------|--------|--------|-------|--------|
|  | -11.590 | -5.215 | -0.240 | 4.348 | 11.335 |

```
##
## Coefficients:
```

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -3.6851  | 5.0333     | -0.732  | 0.474        |
| lectures    | 1.8250   | 0.1418     | 12.869  | 1.62e-10 *** |

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.868 on 18 degrees of freedom
## Multiple R-squared:  0.902, Adjusted R-squared:  0.8965
## F-statistic: 165.6 on 1 and 18 DF, p-value: 1.625e-10
```

# Model Checking

When fitting linear regression models there are some assumptions that we make about the relationship we are modelling.

- 1 We assume that the relationship between the dependent and independent variable is linear.
- 2 We assume that the errors in our model are normally distributed with a mean of 0. ( $\epsilon_i \sim N(0, \sigma^2)$ ).
- 3 We assume errors have constant variance (no heteroscedasticity)
- 4 We assume errors are independent (ie. there is no pattern in the residuals).

We need to check that the models we fit do not violate any of these assumptions because if they do then linear regression is not an appropriate model for the data.

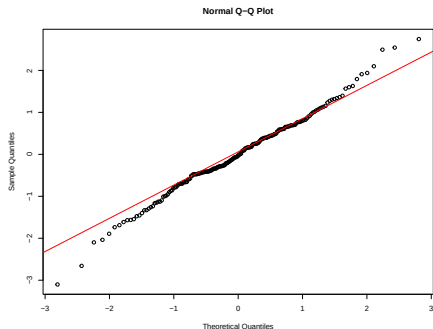
# Assessing linearity

To ensure that the relationship we are modelling is linear we can just plot a scatter plot of  $x$  and  $y$  and look if it looks linear.

# Assessing normality of residuals

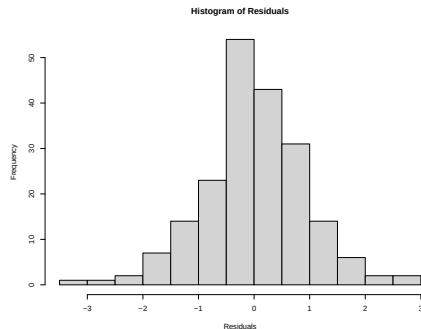
There are two ways to check this assumption (you should always do both).

## 1. QQ-plot



We want the points to follow the red line (a bit of deviation in the tails is alright).

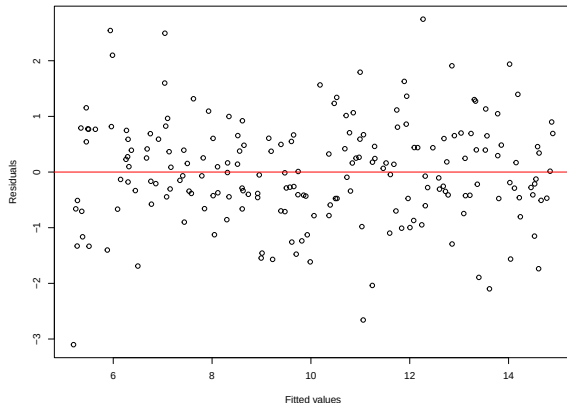
## 2. Histogram



We want the histogram to be bell-shaped (should look normally distributed).

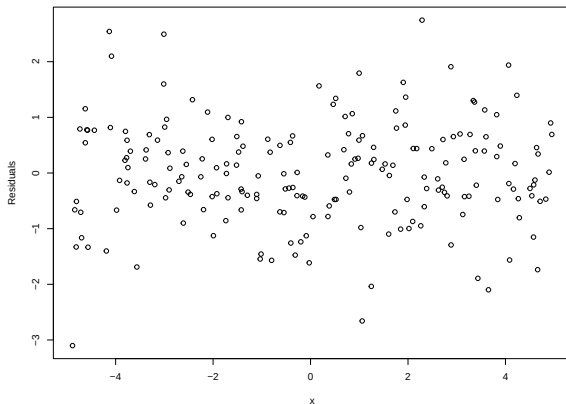
# Assessing constant variance of errors

We can check this assumption by plotting the residuals versus the fitted values. If there is no pattern (just random scatter) then this assumption is met.

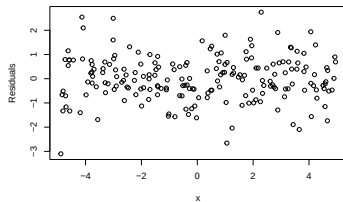
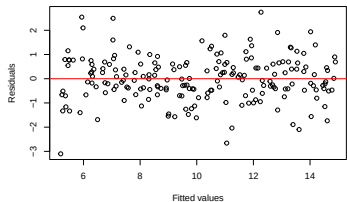
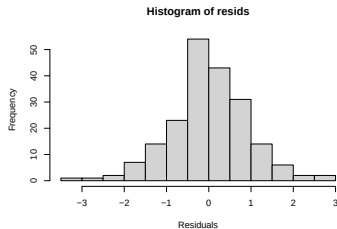
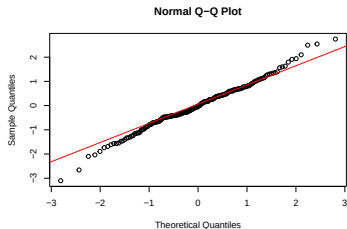


# Assessing independence of errors

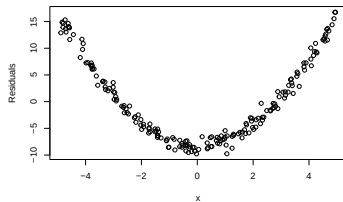
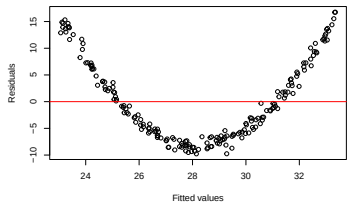
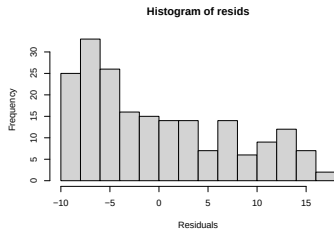
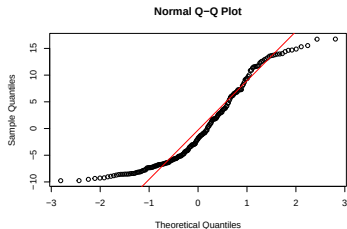
We can check this assumption by plotting errors against the independent variable. If there is no pattern then the errors are independent (don't change with  $x$ ).



# An example of a model that meets assumptions



# An example of a model that violates assumptions



# Prediction

Linear regression can be used to quantify the linear relationship between two variables but it can also be used to predict the value of the dependent variable from the value of the independent variable.

- The predicted values from a linear regression lie on the fitted line (ie. if you supply an  $x$  value, the predicted  $y$  value is that of the corresponding point on the fitted line.)
- Prediction should only be done once you have confirmed that your model fits the data well and has passed all of the model checks.
- You cannot predict outside of the range of your data (for example, if you only have  $x$  values ranging from 0 to 10, you can't predict for an  $x$  of 15).

# Predicting marks based on lecture attendance

Equation for a fitted line:

$$\hat{y} = \beta_0 + \beta_1 x$$

For our example the fitted line is given by:

$$\begin{aligned}\widehat{\text{mark}} &= \beta_0 + \beta_1 \times \text{lectures} \\ \widehat{\text{mark}} &= -3.6851 + 1.825 \times \text{lectures}\end{aligned}$$

If we want to know the predicted mark of a student who attends 35 lectures for the course we can do the following:

$$\begin{aligned}\widehat{\text{mark}} &= -3.6851 + 1.825 \times 35 \\ \widehat{\text{mark}} &= 60.19\end{aligned}$$

So a student who attends 35 lectures can expect a mark of 60.19% for the course.

# Confidence and Prediction interval

We saw that we can use a confidence interval to provide a level of confidence around our  $\beta$  estimates. We can also get a level of confidence around our predictions.

We can predict the average  $y$  value for a given  $x$  value or we can predict the  $y$  value for a specific individual given their  $x$  value. These actual predictions will be the same if the  $x$  value is the same but the intervals will differ.

**CI for average  $y$ :**

$$\hat{y} \pm t_{\frac{\alpha}{2}, n-2} s_{\epsilon} \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}$$

**PI for individual  $y$ :**

$$\hat{y} \pm t_{\frac{\alpha}{2}, n-2} s_{\epsilon} \sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}$$

The CI for the average  $y$  will always be narrower than the PI for an individual  $y$ . This is because predicting for an individual is always more uncertain than predicting the mean.

# Prediction in R

```
#-----
#Predicting for a single individual
#-----

indi_lectures <- 35
predict(fit, newdata = list(lectures=indi_lectures), interval = "prediction")
```

```
##          fit          lwr          upr
## 1 60.19001 45.40081 74.9792
```

```
#-----
#Predicting over lecture range
#-----

lectures_new <- data.frame(lectures = 10:48)
predict(fit, newdata=lectures_new)
```

```
##          1          2          3          4          5          6          7          8
## 14.56490 16.38990 18.21491 20.03991 21.86492 23.68992 25.51492 27.33993
##          9         10         11         12         13         14         15         16
## 29.16493 30.98994 32.81494 34.63995 36.46495 38.28995 40.11496 41.93996
##         17         18         19         20         21         22         23         24
## 43.76497 45.58997 47.41498 49.23998 51.06498 52.88999 54.71499 56.54000
##         25         26         27         28         29         30         31         32
## 58.36500 60.19001 62.01501 63.84001 65.66502 67.49002 69.31503 71.14003
##         33         34         35         36         37         38         39
## 72.96503 74.79004 76.61504 78.44005 80.26505 82.09006 83.91506
```

# A note on causality

It is important to remember that there being correlation between two variables **does NOT** imply that one variable causes a change in the other variable.

We can identify that there is a relationship but we can't determine directionality or causality in the relationship (for example, there could be a third unmeasured variable that affects both variables creating the observed relationship, in this case neither variable causes the other).